

Video Transcript

Video 6: The MBTA API (03:38)

When it comes to cities, the MBTA API, is a pretty friendly one. You can navigate to `api-v3.mbtta.com`, and register as a developer, and get a key. But for our purposes, experimentation, we can really go without any keys as you can see here. You can use the following URL to go directly without registering, and use the following path to pull back the data of route 1. You can see there, that the very first part of the URL refers to vehicles, the MBTA also has trains, ferries, and subways. And that, in this case, we don't just want all vehicles; we want to filter to route 1 and we want to include the trip.

So, let's take a look at this address in the browser. Let's paste it on the address bar and as you can see there, we have a list of buses in its data. This is live at this specific second, the information that is coming back. You should also note that in case you are on time zones where you might have the API at times that no buses are running, in which case you will not get any data because there are no buses on the road. So, take that into account.

Now, if we were to search here for latitude and longitude, you can see here that we would start to get those for a number of buses, if we scroll up and down. Now you could also see that there is some information, for example, about seating. You can see there are a few seats available, many seats available, and so on, for the data. So, as you can see there, we can go directly to that address and pull back the information for that route.

And so, the next thing that we're going to do is that we're going to write a very simple client that is simply going to hit that endpoint, and bring back that data and write it to the console. As you can see here, this is the simple program. It has the address or the URL that we just used. It then pulls it back using `request.urlopen`. Since this is JSON data, it parses it using `json.loads`, and then we're simply going to write that to the console. So, here I have an empty file that I've just created, it's called `mbta_client.py`. Into it, I'm going to paste the URL information and I'm going to uncomment this part here, and I see what happened.

My syntax got broken up there from doing the copy and paste from PowerPoint. And now, I have

it within the file, I'm going to go ahead and save it. I'll go ahead and open up the terminal. As you can see there, I'm going to go ahead and move down a little bit so we can see all of our code. I'm then going to go ahead and enter `python3`, the name of my file, and let's go ahead and run it. And as you can see there, we got the exact same data that we did. Well, it's not the exact same one because time has gone by and the buses most likely have moved.

But we had the same behavior that we did when we accessed through the browser using the address. But now, we're doing that programmatically, and you can see there, that we're parsing that information there as well. So, you can use that same approach on your server when you are polling and requesting information from MBTA.

